

CS201 Lab 6

1. Consider a body moving under gravitational (attractive inverse-squared) force, $V(r) = -\frac{4\pi^2}{r}$.
 - (a) Numerically integrate (by Euler's method) the equations of motion and demonstrate orbits of different shapes (circle, ellipse etc.). Mention clearly what parameters you adjust to obtain the different trajectories and why. Investigate the dependence on step-size of Euler's method on the results by displaying the results for at least one closed trajectory for three (suitable) different step-sizes. Comment on your results.
 - (b) Plot the value of the energy at each step of Euler's method. Comment on the results.

Remember that you do not necessarily need to cast the equations of motion in Cartesian form (although you may do it if you wish). Only the graphical results are to be displayed in Cartesian coordinates.